

Passport Layout Analysis for OCR

1 Passport Layout Variations

Passports are internationally standardized travel documents used for identity verification and border control. Although passport layouts evolve over time, their machine-readable structure follows the ICAO Doc 9303 standard. OCR systems must accurately extract textual fields, photographs, security features, and the Machine Readable Zone (MRZ) under diverse acquisition conditions.

1.1 Passport Cover



Figure 1: Republic of India Passport cover used for document classification.

The passport cover enables document classification and nationality detection before OCR is applied to the bio-data page.

1.2 Legacy Indian Passport



Figure 2: Legacy Indian passport bio-data page.

Older passports differ in typography, security background, photograph placement and MRZ positioning, making them important for OCR benchmarking.

1.3 Current Indian Passport



Figure 3: Current Indian passport bio-data page.

The current passport layout follows ICAO standards while incorporating modern security printing and standardized field placement.

1.4 Indian e-Passport

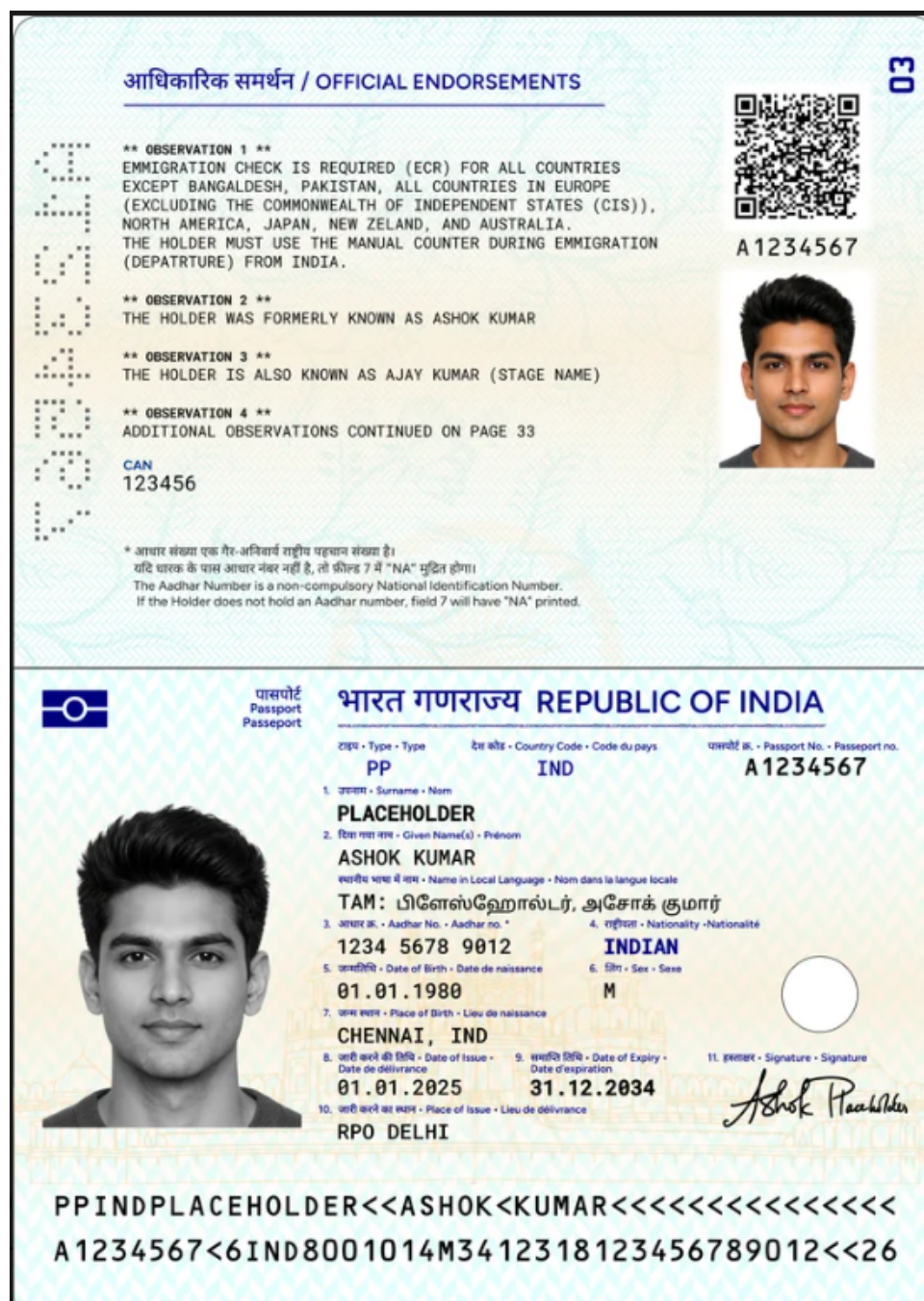


Figure 4: Latest Indian e-Passport design with enhanced security features.

The e-Passport integrates additional digital security features and improved document design while maintaining compatibility with automated passport processing systems.

1.5 Machine Readable Zone (MRZ)

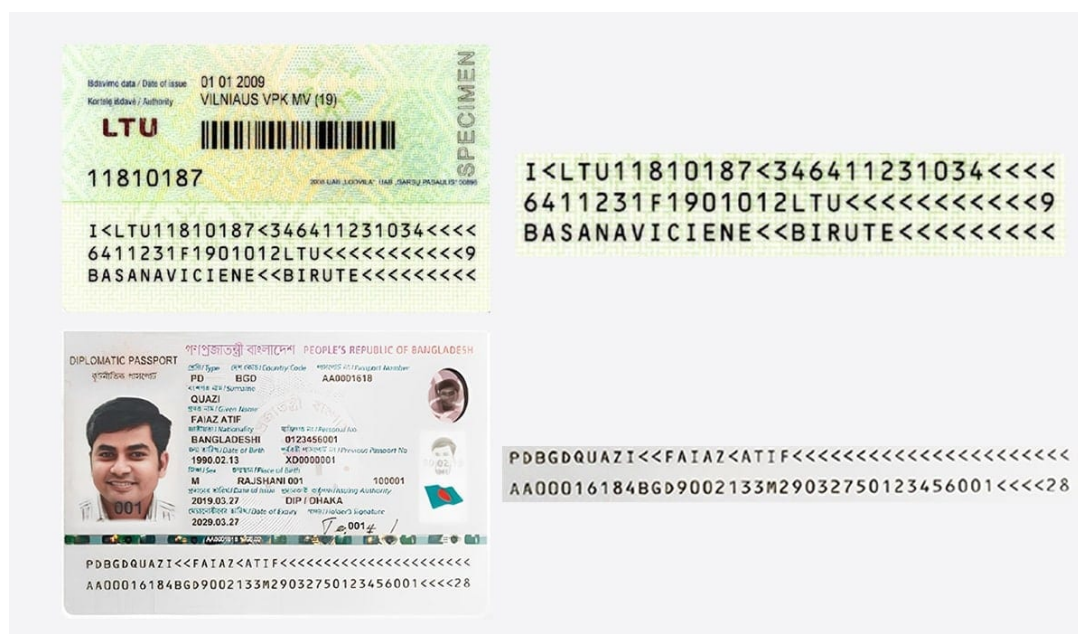


Figure 5: Examples of Machine Readable Zone (MRZ) formats.

The Machine Readable Zone (MRZ) is located at the bottom of the passport bio-data page and follows the ICAO Doc 9303 specification. It encodes the passport number, issuing country, holder's name, nationality, date of birth, sex, expiration date and checksum digits in OCR-B format, enabling automated passport verification.

2 Passport Document Structure

Table 1: Primary Passport Identifiers

Identifier	Description
Passport Number	Unique passport identifier
Surname	Family name of the holder
Given Name	Holder's given name
Nationality	Country of citizenship
Date of Birth	Holder's date of birth
Sex	Gender (M/F/X)
Place of Birth	Birth location
Date of Issue	Passport issue date
Date of Expiry	Passport expiry date
Issuing Authority	Passport issuing office
Photograph	Holder photograph
Signature	Holder signature
Machine Readable Zone	ICAO standardized OCR-B text

3 Identifiers and OCR Keywords

Common textual anchors used during passport OCR include:

- Republic of India
- Passport
- Passport No.
- Type
- Country Code
- Passport Number
- Surname
- Given Name
- Nationality
- Date of Birth
- Place of Birth
- Date of Issue
- Date of Expiry
- Authority
- Sex
- Machine Readable Zone (MRZ)
- ICAO
- P<IND

4 OCR Challenges

Passport OCR involves several challenges due to security printing and varying capture conditions.

- Multiple passport layouts
- Legacy and current passport formats
- Fine security background patterns
- OCR-B font recognition
- Machine Readable Zone extraction

- Checksum validation of MRZ fields
- Holograms and watermarks
- Small font sizes
- Perspective distortion
- Motion blur
- Shadows and reflections
- Camera-captured document images

Table 2: Relative OCR Difficulty Across Passport Layouts

Layout	OCR Difficulty
Passport Cover	Low
Legacy Passport	Medium
Current Passport	Medium
e-Passport	Medium
Machine Readable Zone	High